

rectilinear steiner tree puzzle

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with thanks to Stefan Hougardy (U Bonn)

and Zac Friggstad (U Alberta)

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Stefan Hougardy works at

Bonn Institute for Discrete Math

shares building with

Arithmeum



<https://www.arithmeum.uni-bonn.de>



<https://www.arithmeum.uni-bonn.de/en/visitor-information/shop/details>

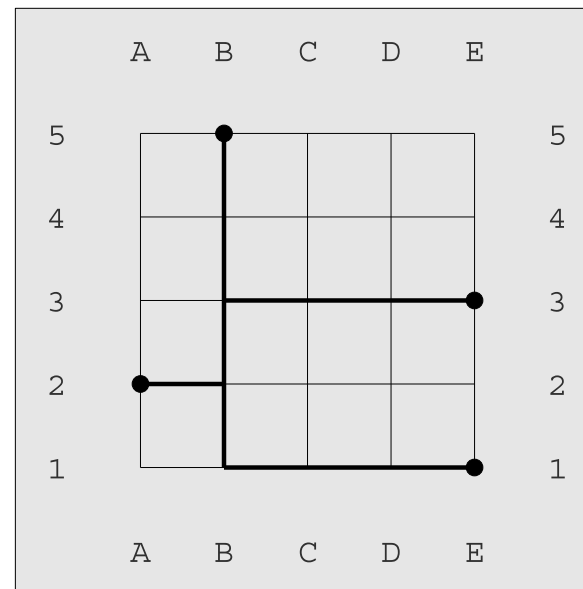
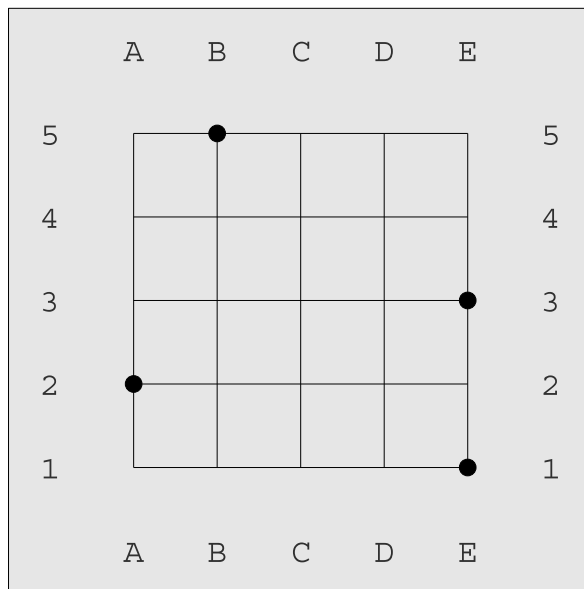
rectilinear steiner tree puzzle:

join all given pins with

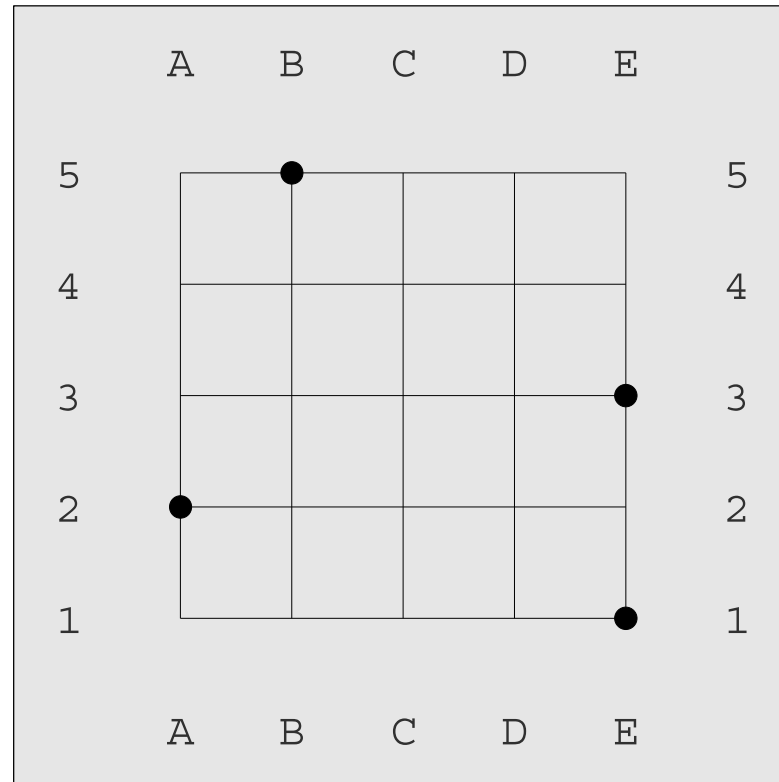
horizontal/vertical wires,

tree *cost* is sum of lengths of wires.

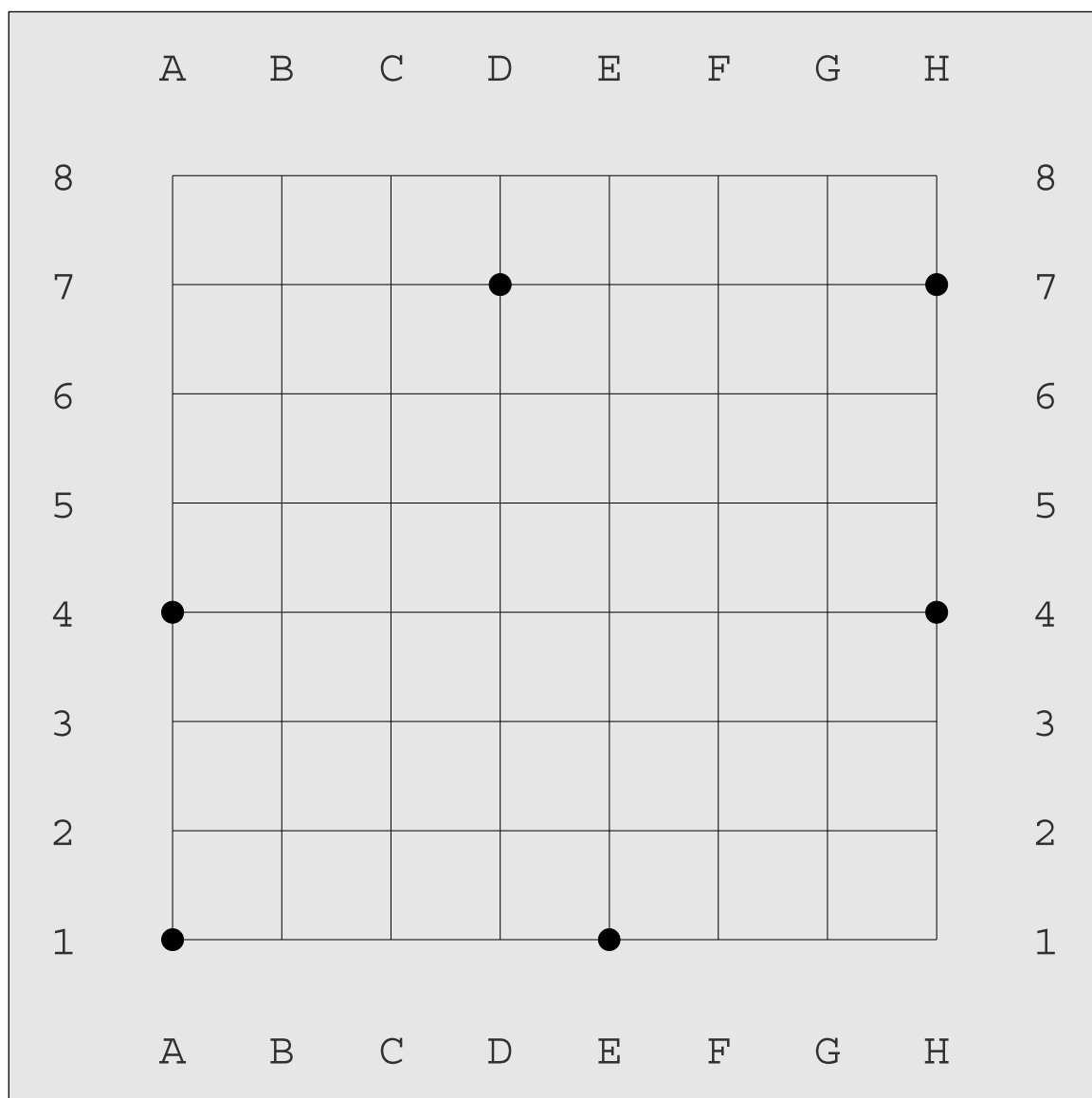
this tree has cost 11, can you do better?



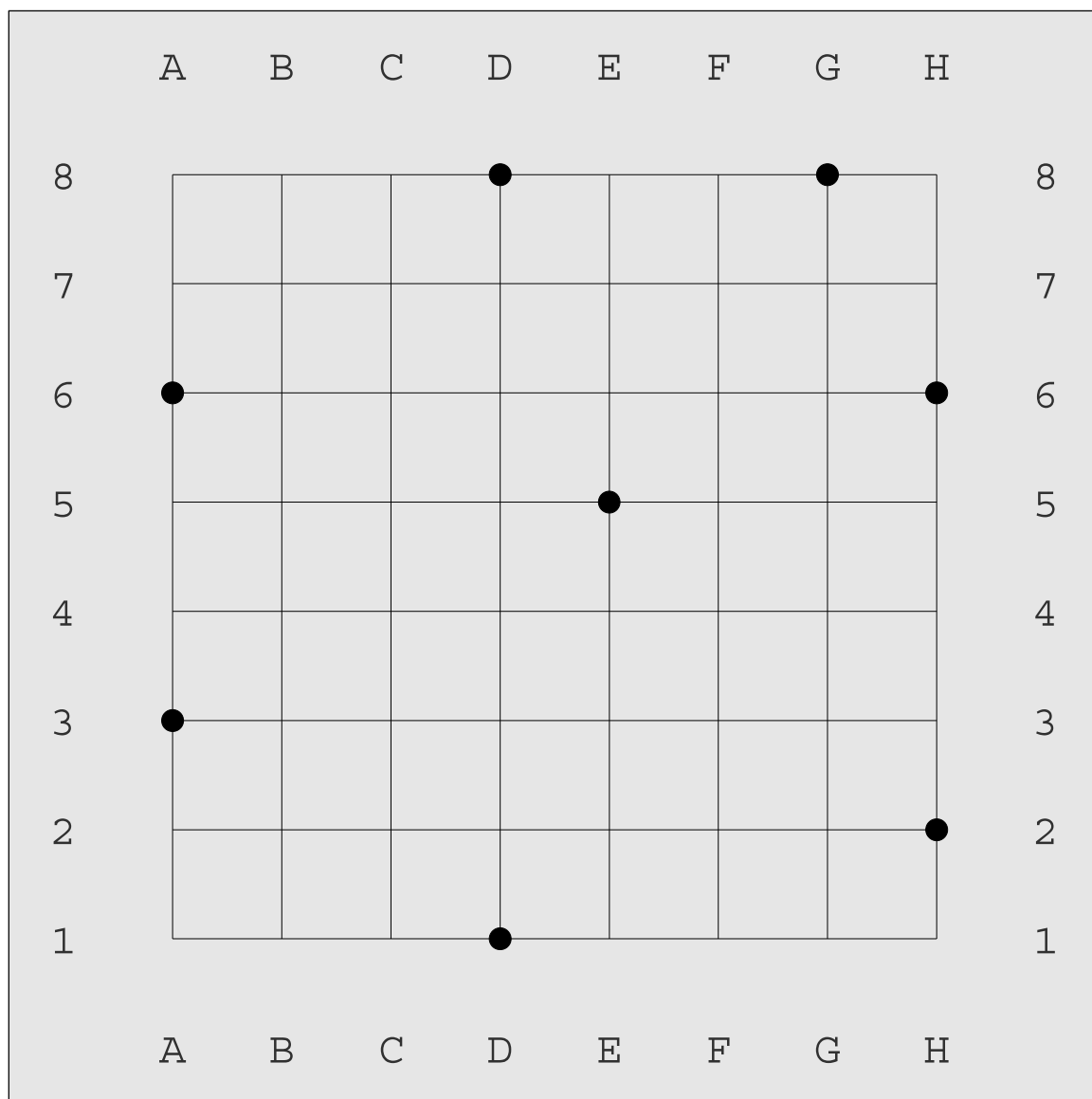
for each puzzle, find a min-cost rectilinear steiner tree



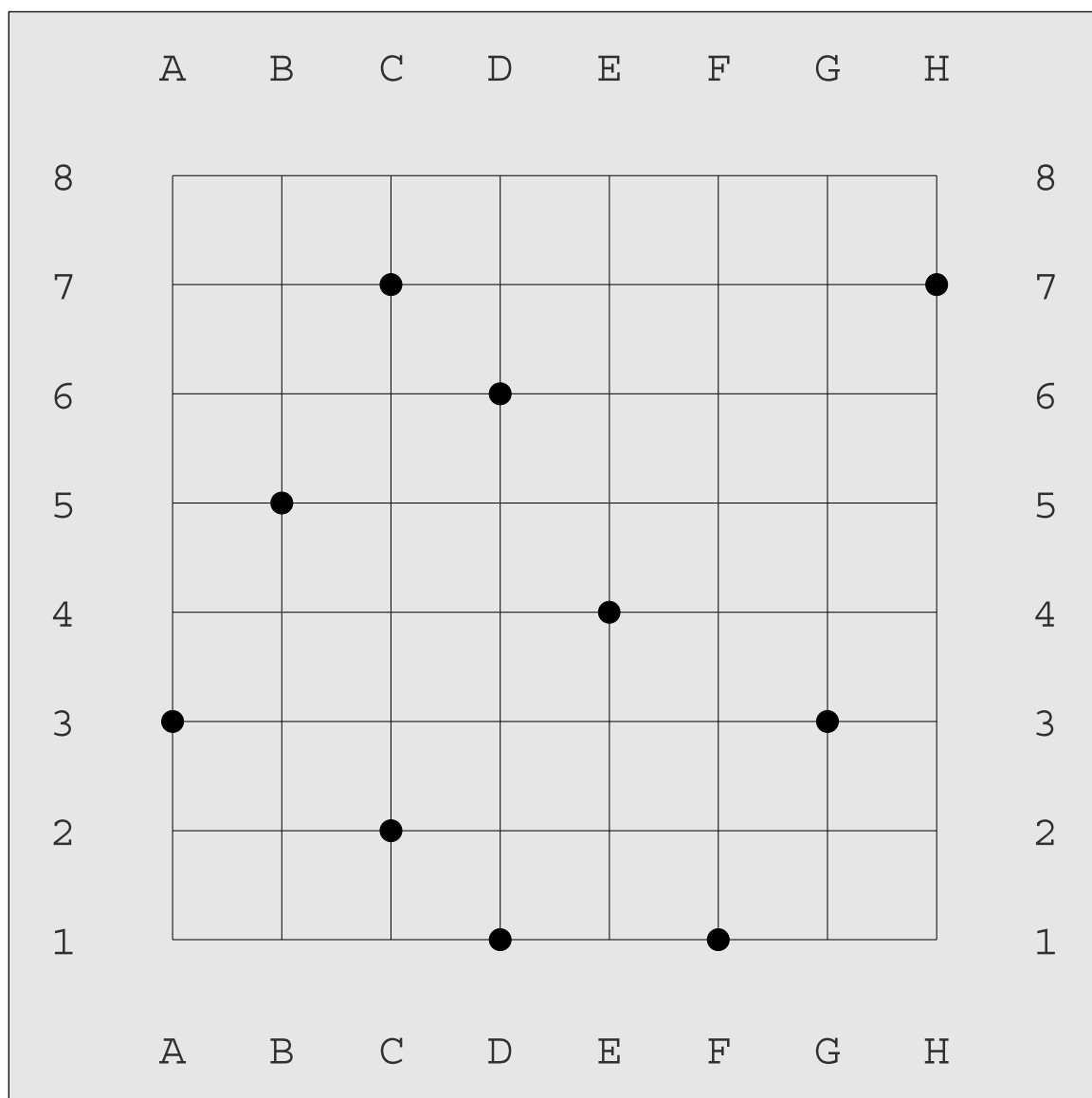
4 pins, cost 9



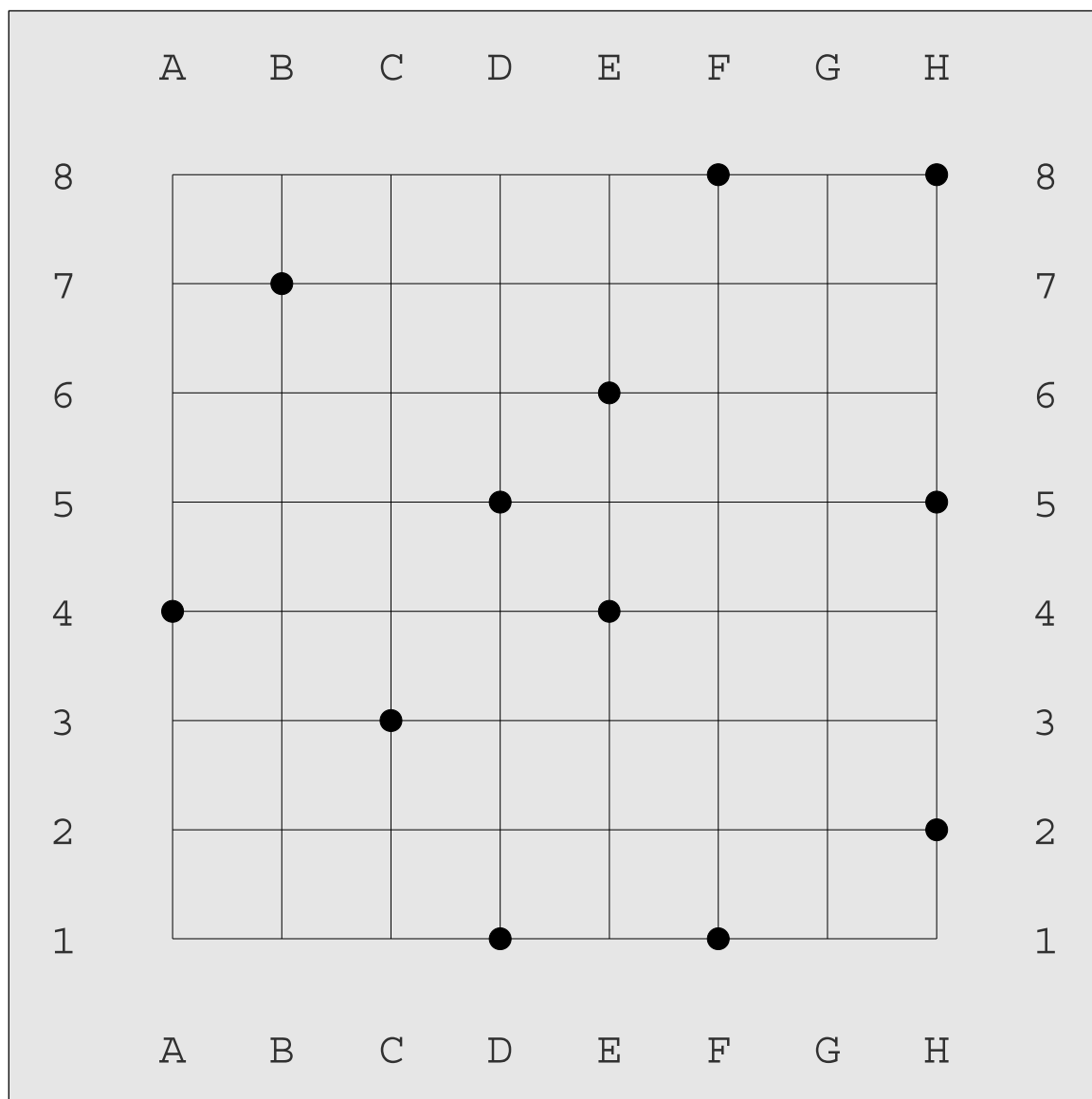
6 pins, cost 19



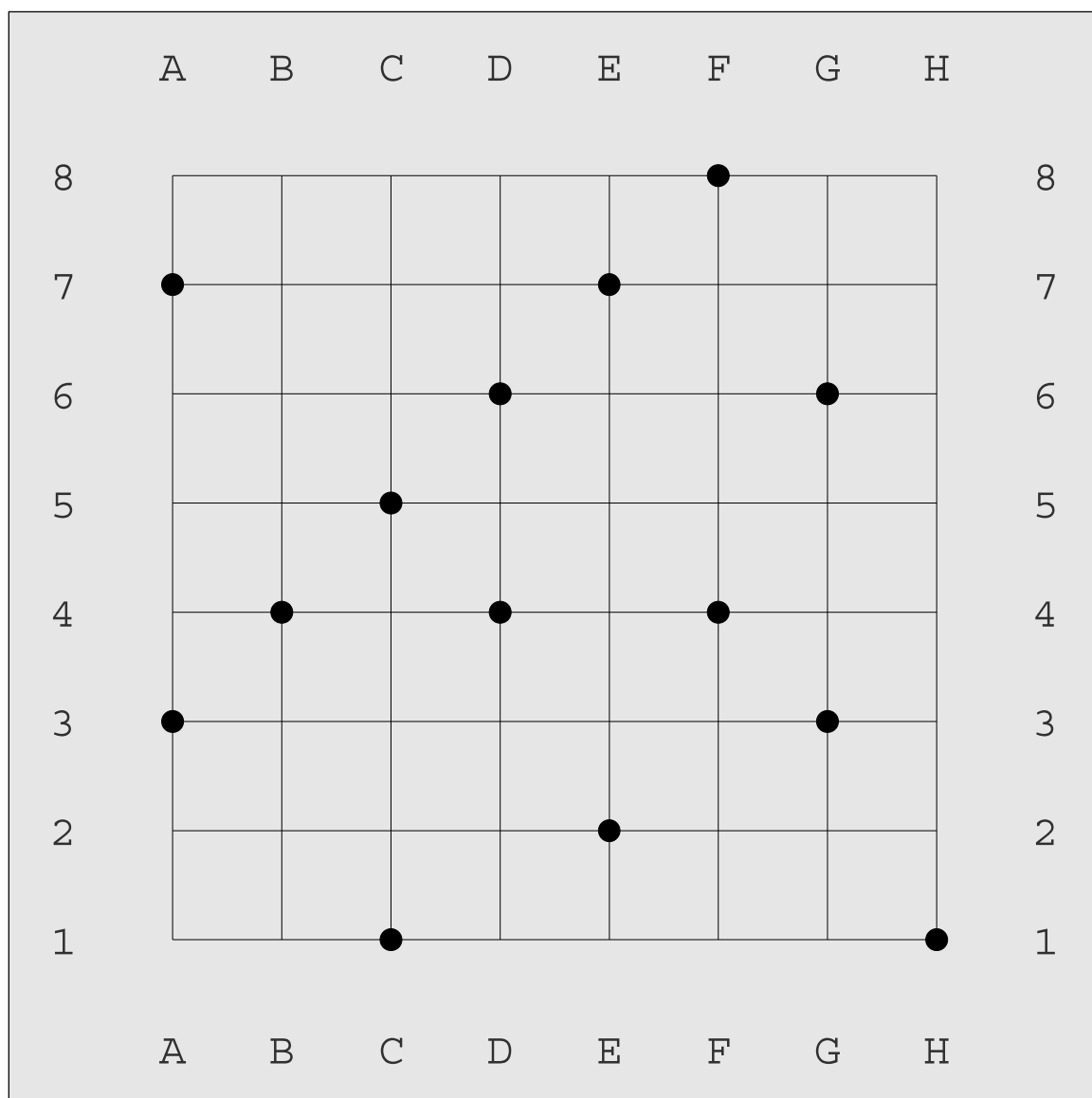
8 pins, cost 23



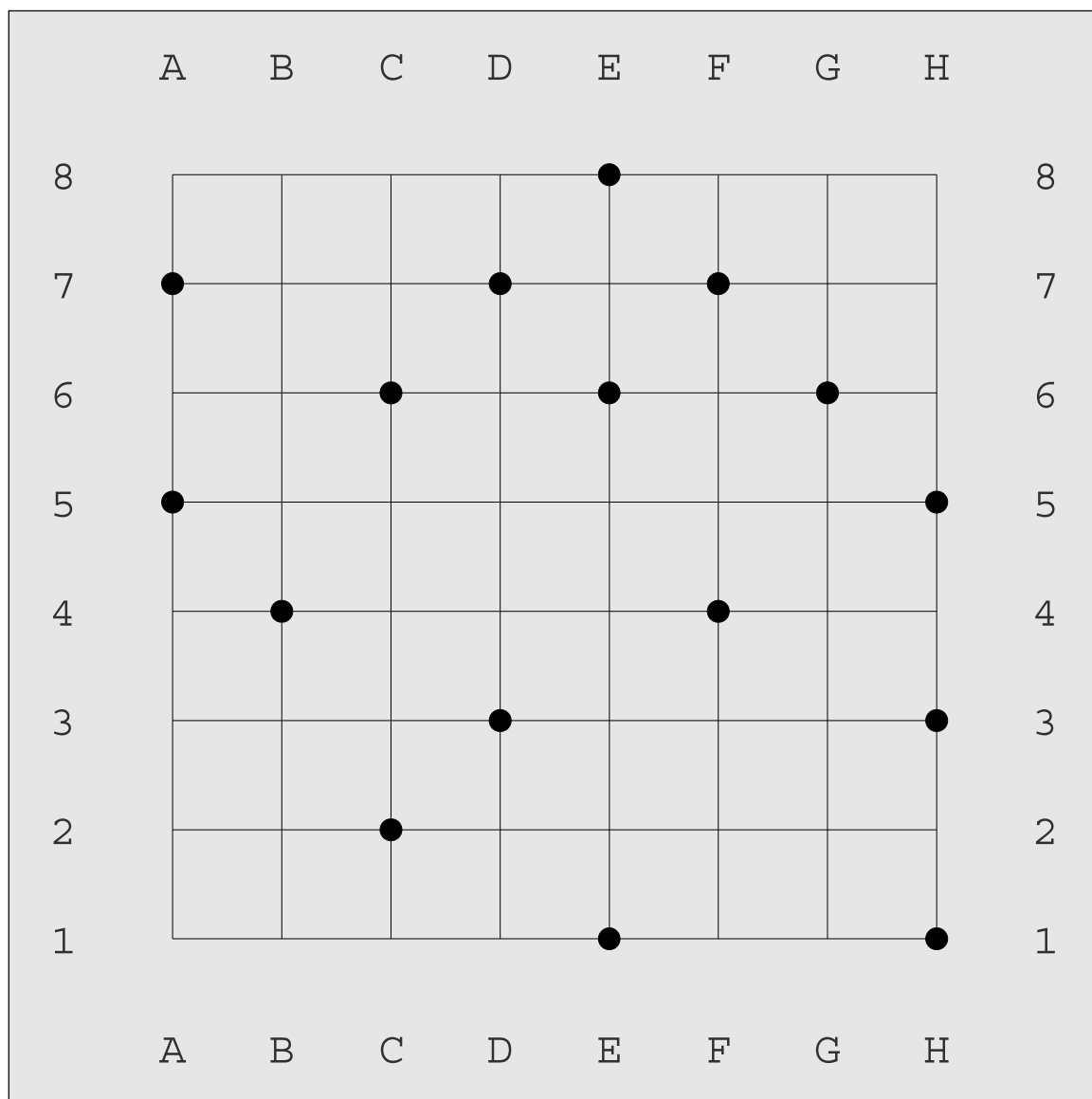
10 pins, cost 22



(harder) 12 pins, cost 26



(harder) 14 pins, cost 27



(harder) 16 pins, cost 28

thank you

steiner tree approx

- from input G , create pins-graph $P(G)$
- $P(G)$ node set: pins of G
- $P(G)$ edge set: all pairs of pins of G
- for each edge of $P(G)$, cost is min-cost path in G
- find mst M of $P(G)$
- claim: G has steiner tree with cost $\leq \text{cost}(M)$
- proof:
- for each edge (x, y) of M , add corresponding min-cost path in G
(on the path, don't add any edge that would create a cycle)

